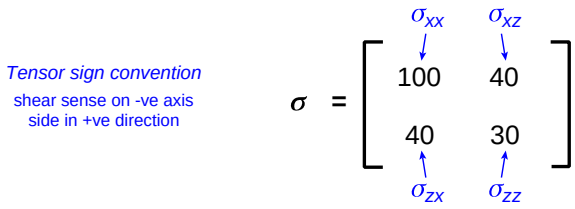
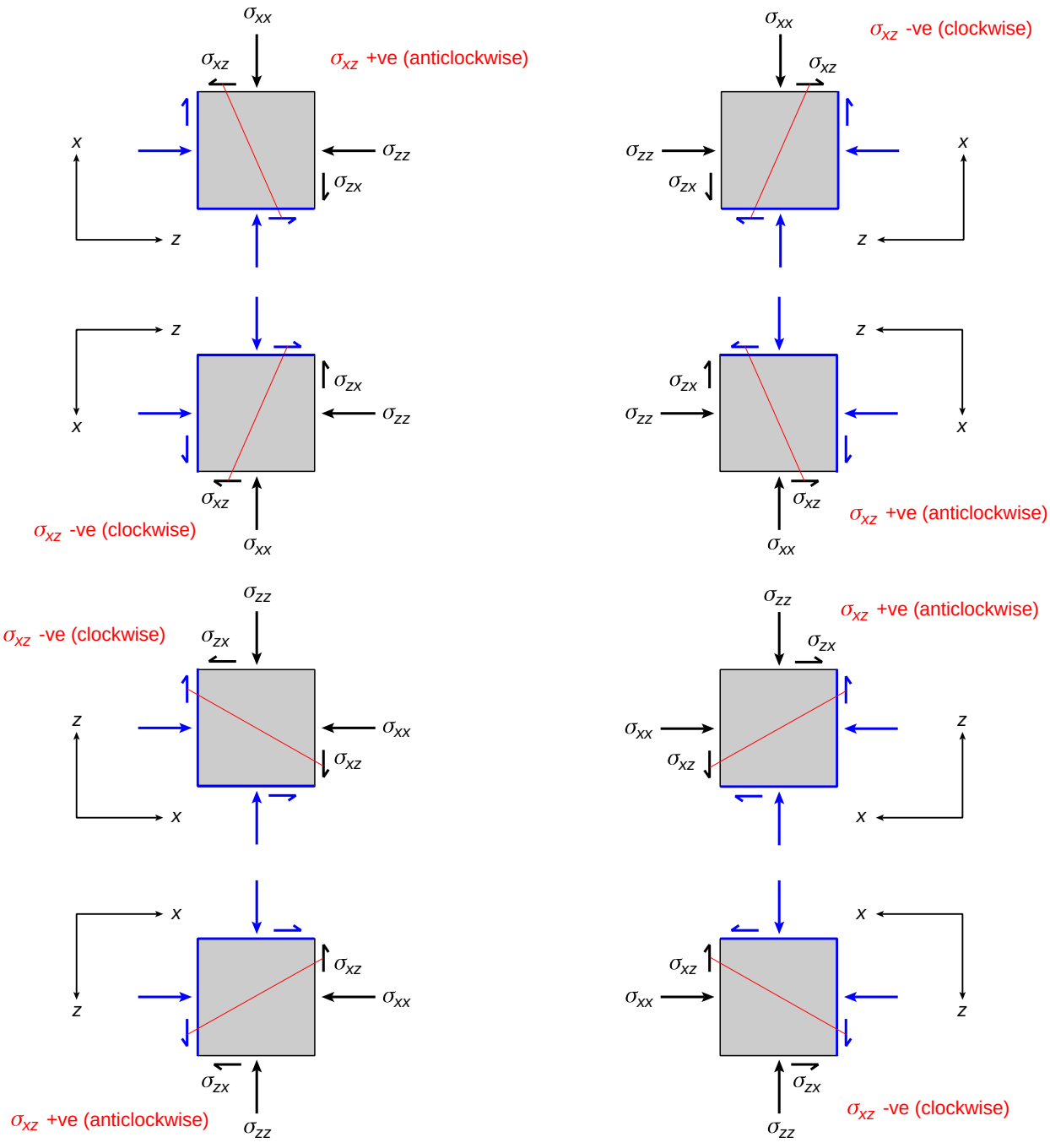
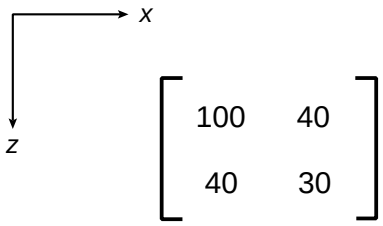
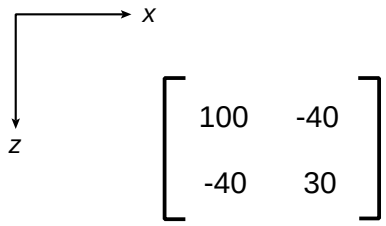
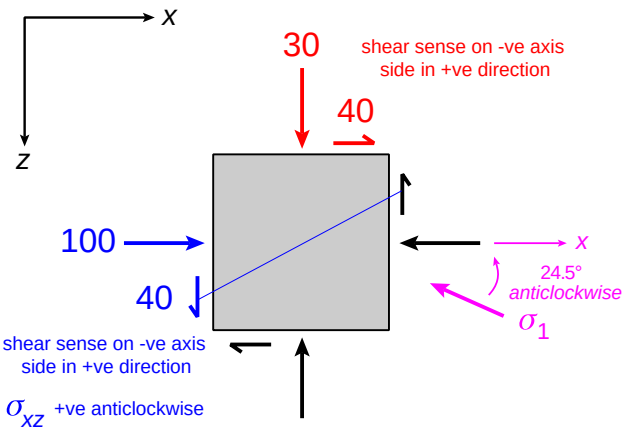
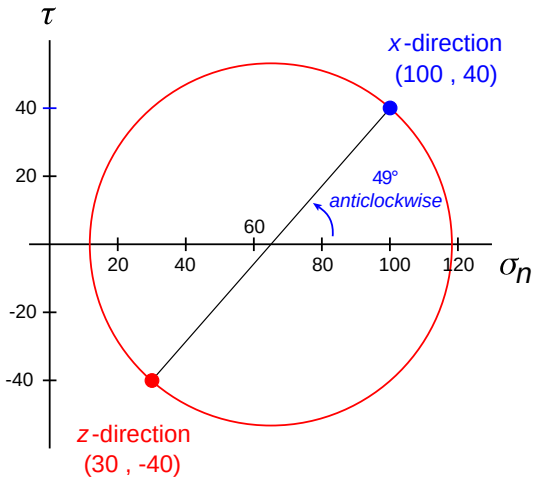
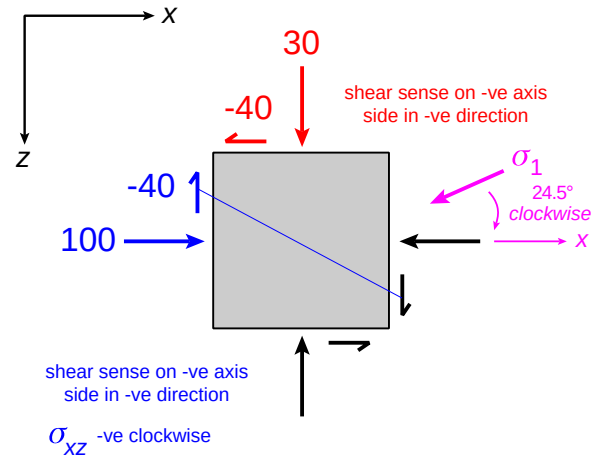
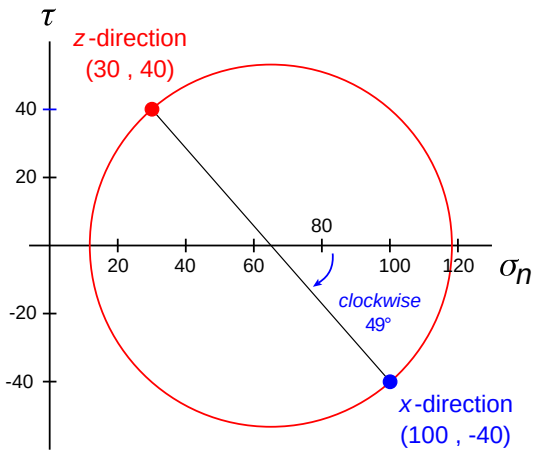
# Exercise 2 solution

A state of stress (units MPa) is given in a Cartesian coordinate frame $(x, z)$ by

$$\begin{bmatrix} 100 & 40 \\ 40 & 30 \end{bmatrix}$$

(a) Draw arrows showing the sense of $\sigma_n$ and $\tau$ on each of the squares below given the indicated orientation of the coordinate frame. In each case, determine the sign of $\sigma_{xz}$ using the Mohr sign convention.






(b) Draw Mohr circles for the following two states of stress, and indicate, on a square element, the direction of $\sigma_1$.